# INSTRUCTOR KEY: RAY KESSLER TEAM CASE STUDY

**about 2 hours**

## What this is

A simple 2-hour team case study. The whole class (designed for 7 students) works as ONE team and cares for Ray Kessler (74, hypermagnesemia with kidney failure) across 24 hours of patient time. The class advances **4 hours together** (0700, 1100, 1500, 1900, 2300, 0300, then a final check at 0700). Each team fills in **one plain page per checkpoint**. The packet is 12 pages. This key has a run of show, a short model answer for every box, a scoring guide (20 points), a group score sheet and debrief questions.

## Roles and pairs (7 students, one team)

- **Charge nurse** (rotates; each student has exactly one stop): runs the section, keeps time, **asks each pair what they found and what to do**, decides the priorities, makes the SBAR call and is the only one who writes in the shared boxes.
- **Pair 1, Meds & Treatments (A, B):** MAR, orders, notes: what is running, due or should stop; give or hold meds.
- **Pair 2, Vitals & Assessment (C, D):** read your bedside monitor setting, chart the vitals, report the bedside exam (alertness, reflexes, breathing, urine).
- **Pair 3, Results & Labs (E, F):** new labs, ECG, trends against the diagnosis, the Mg strip.
- **G, float nurse:** helps wherever the charge nurse sends them (for example, runs the clock or marks the strip). G is charge at the final 0700.
- Charge order: A (0700), C (1100), E (1500), B (1900), D (2300), F (0300), G (final). The pair job continues when one of the pair is charge: the **partner reports for the pair**.

## Pairing suggestions

- The rotation is fixed so everyone leads once. Within each pair, match a confident reader with a quieter student.
- Seat the pairs together at the table, with the charge nurse at the head holding the shared packet. Give the Assessment pair the bedside monitor.

## Before class

- Print the shared packet single-sided: 2 or 3 copies for the table (or project the pages). Only the current charge nurse writes in the shared boxes while the pairs advise. **Print 7 copies of page 12** (individual reflection and peer check), one per student. Have the key open and the EHR synced to patient time Tue 0700 (5E-14, SIM-40318).
- Have the monitor ready to set to the values in the **Monitor to set** row of each checkpoint. Students read it and chart it in the NOW row.
- Show the team strip on page 2. It is empty on purpose: students mark each magnesium result themselves.

## Rules that protect the pacing

- Students finish a page at the **end** of its time block. Nothing on a later page is printed on an earlier one. Do not let anyone turn the page until you advance the clock.
- Every number in this key matches the patient chart (patient.json) and the monitor. If the team and key disagree, check the chart.
- Acceptable answers can differ from the model. Look for chart numbers and a reason, not exact wording.

# Run of show (about 2 hours)

Each checkpoint follows the same 14-minute rhythm (the 1500 gets 17 because it has a second page).

| Clock | Minutes | What the instructor does |
|---|---|---|
| Start | 0:00-0:05 | Welcome. Everyone sits together as one team; fill in the pair names (A to F) and the float nurse (G). Walk through the cover: roles, rotation table, rubric (page 3) and the Mg strip (page 2). Student A is charge nurse at 0700. Remind the charge nurse to ask each pair for what they found before deciding. |
| 0700 | 0:05-0:19 | Advance the clock and set the monitor. Read the situation aloud. Charge A runs the page (B reports for pair 1): asks each pair in turn (1-3), pairs look and report while the charge nurse writes (3-10), the charge nurse makes the SBAR call to you (10-13) and you answer as the provider. Move to the next clock. |
| 1100 | 0:19-0:33 | Same rhythm. New charge nurse (C); D reports for pair 2. Watch for skipping the call for help. |
| 1500 | 0:33-0:50 | Same rhythm, 17 minutes: the second page covers what must be ready for dialysis, what to watch, and the call numbers. Charge nurse E; F reports for pair 3. |
| 1900 | 0:50-1:04 | Same rhythm. Charge nurse B; A reports for pair 1. Watch for the low glucose. |
| 2300 | 1:04-1:18 | Same rhythm. Charge nurse D; C reports for pair 2. Watch for who thinks of the sleep-aid order. |
| 0300 | 1:18-1:32 | Same rhythm. Charge nurse F; E reports for pair 3. Watch for who refuses the "pink milk". |
| Final 0700 | 1:32-1:44 | Advance, set the monitor, read the situation. The team completes the final page (charge nurse G) (priorities for the day team, a handoff SBAR, teaching tips, the story of the stay). |
| Reflection | 1:44-1:50 | Each student completes their own reflection and peer check (page 12). Collect the shared packet and the 7 reflections. |
| Debrief | 1:50-2:00 | Use the debrief questions at the end of this key. Show the finished strips. |

## The same 14 minutes, one checkpoint

| Minute | What happens |
|---|---|
| +0 | Advance the clock (the 4h button). Set the monitor. Read the situation aloud. Remind them that the charge nurse changed. |
| +1 to +3 | Charge nurse asks pair 1 (meds), pair 2 (assessment and vitals), pair 3 (results and labs) in turn; each pair reports and the charge nurse writes. (Do not tell them the answers.) |
| +3 to +10 | Nurses look at the chart and the monitor and write. Walk around. Ask a charge nurse "what is your team deciding?" |
| +10 to +13 | Charge nurse makes the SBAR call to you. Answer as the provider with the suggested reply in this key. Keep it short. |
| +13 to +14 | Check that the magnesium level is on the strip and the prediction is written. Then advance the clock. |

# 0700 Tuesday: Checkpoint 1 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 52, BP 96/56 (MAP 69), RR 14, SpO2 94% room air, temp 97.6, pain 2/10 low back, glucose 132; rhythm: Sinus bradycardia with first-degree AV block (PR 0.26 s)

**Urine for the vitals box:** 130 mL since the Foley went in at 0610 (ED block 0430-0700); the plan for 0700-0800 is 35 mL

## New results (chart values)

| Test | Normal | Tue 0440 |
|---|---|---|
| Magnesium | 1.7-2.2 | 7.8 |
| Potassium | 3.5-5.1 | 5.4 |
| Creatinine | 0.7-1.3 | 4.9 |
| BUN | 7-20 | 84 |
| Calcium | 8.6-10.2 | 8.1 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | Due: second NS 500 mL bolus at 0715 (listen to the lungs before and after), then NS 100 mL/h; furosemide 80 mg IV at 0915; calcium gluconate if needed. Stop or hold: every magnesium product from the bag (milk of magnesia, Maalox-type antacid, magnesium oxide), ibuprofen and lisinopril. Check that no order or bowel protocol adds magnesium. |
| **VITALS & ASSESSMENT pair** | HR 52 with first-degree AV block (PR 0.26 s), BP 96/56, RR 14, SpO2 94% on room air. Drowsy, slurred speech, reflexes only 1+, weak grip, flushed face, dry mouth, skin tents. Urine dark: 130 mL since the Foley (0610). |
| **RESULTS & LABS pair: new results and significance** | Magnesium 7.8 (critical; normal 1.7-2.2), potassium 5.4, creatinine 4.9 (usual 2.6), BUN 84, calcium 8.1. The kidneys cannot get rid of magnesium: long-term kidney disease plus a sudden drop from vomiting, dryness and kidney-hurting medicines, plus three magnesium products taken at home. |
| **RESULTS & LABS pair: Mg level and strip** | Write 7.8 mg/dL; mark on the strip in the Reflexes lost band (7-10). |
| **TEAM (charge): what changed** | First checkpoint, so compare with the ED (0425): HR 46 to 52 and BP 82/46 to 96/56, a little better after the bolus. Still drowsy and slow. |
| **TEAM (charge): top 3 priorities** | 1) Airway, breathing, circulation and nerve checks: full-minute RR, SpO2, monitor on, head up 30 degrees, suction at the bed, nothing by mouth until a swallow screen. 2) Find and stop every magnesium source: ask Eleanor, review the bag, tell the provider and pharmacy. 3) Safe fluids and kidneys: lung check before and after the bolus, hourly urine, hold the home medicines that hurt the kidneys. |
| **TEAM (charge): SBAR** | **S:** RN for Raymond Kessler, 5E-14. His wife brought a bag of home products that include milk of magnesia, a magnesium antacid and magnesium oxide. **B:** 74, kidney disease with a sudden kidney injury (creatinine 4.9), magnesium 7.8 on arrival, HR 52 with a long PR, BP 96/56, reflexes 1+. **A:** The magnesium comes from these products and his kidneys cannot clear it. **R:** Please stop all of them, have pharmacy review, ask nephrology to see him, and check the bowel protocol for magnesium. |
| **TEAM (charge): what we will do next** | Stop all magnesium; lung check then bolus; hourly urine; NPO until swallow screen; ask nephrology to see him; talk with Eleanor. |
| **Prediction (no wrong answers)** | Any reasoned guess earns credit. Expect: magnesium may still rise because he makes little urine. What happens: 8.7 at 0900 and slower breathing. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** Thank you. Stop all the magnesium products and add them to the allergy/avoid list. I will ask nephrology to see him today. Do the lung check before and after the bolus and call me for crackles or SpO2 under 92%.

# 1100 Tuesday: Checkpoint 2 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 44, BP 84/48 (MAP 60), RR 10, shallow, SpO2 91% room air, temp 97.4, pain 0/10, glucose 104; rhythm: Sinus bradycardia, long PR (0.30 s)

**Urine for the vitals box:** last hour 40 mL (0700-1100 hourly 35, 30, 55, 40 = 160 mL); the guide has the instructor say "about 20 mL/h": accept 20-40 mL/h as low

## New results (chart values)

| Test | Normal | Tue 0440 | Tue 0900 |
|---|---|---|---|
| Magnesium | 1.7-2.2 | 7.8 | 8.7 |
| Potassium | 3.5-5.1 | 5.4 | 5.3 |
| Creatinine | 0.7-1.3 | 4.9 | 4.8 |
| Calcium | 8.6-10.2 | 8.1 | 8.7 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | Given since 0700: second bolus, maintenance saline, furosemide 0915. Needed now: calcium gluconate 1 g IV over 10 minutes on the cardiac monitor through a good large vein; oxygen with capnography (oxygen does not fix slow, weak breathing); ask to stop the maintenance saline until reviewed; no sedatives or opioids. |
| **VITALS & ASSESSMENT pair** | HR 44, BP 84/48 (MAP 60), RR 10 and shallow, SpO2 91% on room air. Reflexes now absent, only wakes to a loud voice. Urine 40 mL last hour. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 8.7 (up from 7.8), potassium 5.3, creatinine 4.8, calcium 8.7. Only magnesium got worse. The kidneys make very little urine (about 30 to 55 mL/h, then 40), so magnesium is not leaving. |
| **RESULTS & LABS pair: Mg level and strip** | Write 8.7 mg/dL; mark on the strip in the Reflexes lost band (7-10). |
| **TEAM (charge): what changed** | Worse: HR 52 to 44, BP 96/56 to 84/48, RR 14 to 10, SpO2 94% to 91%, reflexes 1+ to absent, less alert. Magnesium 7.8 to 8.7. |
| **TEAM (charge): top 3 priorities** | 1) Call the rapid response team now (RR 10, HR 44, BP 84/48, reflexes gone, less alert) and stay with Ray. 2) Airway and breathing: head up, stimulate, oxygen with watching of RR and CO2, suction and bag-valve-mask at the bed, strictly nothing by mouth. 3) Heart and antidote: monitor, 12-lead ECG, pacing pads, calcium gluconate 1 g IV over 10 minutes on the monitor. |
| **TEAM (charge): SBAR** | **S:** RN for Raymond Kessler, 5E-14. I am calling a rapid response: RR 10 and shallow, HR 44, BP 84/48, SpO2 91%, only waking to a loud voice, reflexes gone. **B:** 74, kidney disease, admitted with magnesium 7.8 and given calcium in the ED. Home magnesium products found. The 0900 magnesium was 8.7. Urine low. **A:** I think magnesium is still rising. He has slow breathing and is at risk of heart block. **R:** Please come to the bedside now. I would like calcium gluconate 1 g IV, oxygen, STAT magnesium and ECG, nephrology and ICU outreach on the phone, pacing pads at the bed. |
| **TEAM (charge): what we will do next** | Rapid response now; calcium on the monitor; airway equipment ready; stop the maintenance saline until reviewed; update Eleanor. |
| **Prediction (no wrong answers)** | Calcium helps for about half an hour to an hour but does not remove magnesium. Expect a higher level and the need for more than medicine. What happens: 9.5 at 1115 and 10.9 at 1315. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** I am coming now. Give calcium gluconate 1 g over 10 minutes on the monitor, put him on oxygen with capnography, stop the maintenance saline and get a STAT magnesium and ECG. I will call nephrology and ICU outreach.

# 1500 Tuesday: Checkpoint 3 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 46, BP 88/50 (MAP 63), RR 9, SpO2 92% on 4 L nasal cannula, EtCO2 52, temp 97.2, pain CPOT 0, glucose 88; rhythm: Sinus bradycardia, long PR (0.32 s), wide QRS (0.14 s), tall T waves

**Urine for the vitals box:** last hour 10 mL (1100-1500 hourly 20, 25, 15, 10 = 70 mL)

## New results (chart values)

| Test | Normal | Tue 0900 | Tue 1115 | Tue 1315 |
|---|---|---|---|---|
| Magnesium | 1.7-2.2 | 8.7 | 9.5 | 10.9 |
| Potassium | 3.5-5.1 | 5.3 | 5.6 | 5.8 |
| Creatinine | 0.7-1.3 | 4.8 | 5.0 | 5.3 |
| CO2 (bicarbonate) | 22-29 | 18 | 17 | 16 |

| ABG | Normal | Tue 1130 | Tue 1415 |
|---|---|---|---|
| pH | 7.35-7.45 | 7.24 | 7.20 |
| PaCO2 | 35-45 | 46 | 52 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | Calcium gluconate 1 g at 1120, then 2 g doses (1235 and 1400); NS 100 mL/h stopped at 1415 (fluid overload risk); lisinopril and metoprolol held; no blood pressure medicines during dialysis. Dialysis: 3 hours, low-magnesium bath, no heparin. Stop: any more IV fluid. The neck catheter is for dialysis staff only. |
| **VITALS & ASSESSMENT pair** | RR 9 and falling (14, 10, 9), EtCO2 52 (CO2 building up), SpO2 92% on 4 L, HR 46, BP 88/50, only wakes to a loud voice. New crackles in the lungs. Urine 10 mL last hour. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 9.5 then 10.9 (the peak, critical). Potassium 5.6 then 5.8, creatinine 5.0 then 5.3, CO2 17 then 16. ABG at 1415: pH 7.20, PaCO2 52, so he is not breathing out enough CO2. ECG 1330: HR 38, PR 320 ms, QRS 148 ms, tall T waves. Oxygen cannot fix this; the magnesium has to come down. |
| **RESULTS & LABS pair: Mg level and strip** | Write 9.5 and 10.9 mg/dL; mark on the strip in the Danger band (over 10). |
| **TEAM (charge): what changed** | Worse again: RR 10 to 9, HR 44 to 46 (ECG 38), BP 84/48 to 88/50, now on 4 L oxygen, EtCO2 52, QRS wider, magnesium 8.7 to 10.9, urine only 10 mL/h, new crackles. |
| **TEAM (charge): top 3 priorities** | 1) Airway and breathing: RR, EtCO2, SpO2, head up, suction, bag-valve-mask and airway cart in the room; know the ICU call numbers. 2) Dialysis safety: consent and ID confirmed, catheter for dialysis staff only, BP and HR every 15 minutes, calcium on the cart, blood pressure medicines held. 3) Heart and fluid: calcium gluconate on the monitor, watch the crackles, no extra fluid, strict intake and output. |
| **TEAM (charge): SBAR** | **S:** RR 9, SpO2 92% on 4 L, EtCO2 52, HR 46, BP 88/50, only waking to a loud voice, dialysis about to start. **B:** Magnesium 10.9, potassium 5.8, QRS 148 ms, ABG 7.20 / 52, calcium 2 g due at 1400, dialysis catheter placed 1440, chest x-ray clear. **A:** He is not breathing out enough and may need airway help before dialysis ends. **R:** Please come now or tell me your numbers for transfer. I would like the airway cart in the room, a repeat ABG and a decision about the ICU. |
| **TEAM (charge): what we will do next** | Airway cart in the room; vitals every 15 minutes; calcium ready; no extra fluid; hold BP medicines; call outreach at the agreed numbers. |
| **Prediction (no wrong answers)** | Dialysis runs about 3 hours and pulls magnesium down fast. New problems can follow: low blood pressure, low sugar, low calcium. What happens: 4.6 at 1800, BP dip to 82/46, glucose 54. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** ICU outreach will come to the bedside. Call them for RR under 8, PaCO2 over 60 or pH under 7.15, SpO2 under 90% on 6 L, or any drop in alertness. Keep the airway cart in the room. Repeat the ABG in an hour.

## Page 2 of 1500: dialysis boxes

| Box | Model answer |
|---|---|
| **Meds pair: before dialysis starts** | Consent (Eleanor by phone at 1350, two nurse witnesses); ID check; catheter in and x-ray clear; low-magnesium bath, no heparin; NS stopped; calcium 2 g given; blood pressure medicines held; airway cart and bag-valve-mask at the bed; 1:1 nurse. |

| | |
|---|---|
| **Assessment pair: while dialysis runs** | BP and HR every 15 minutes, breathing and EtCO2, rhythm, alertness, the catheter dressing (dry, no bleeding), lung sounds. Keep calcium on the cart. Never flush the catheter, draw labs from it or give drugs through it. |
| **Results pair: when do we call for help?** | Call ICU outreach for any one of: RR under 8, PaCO2 over 60 or pH under 7.15, SpO2 under 90% on 6 L, drop in alertness, loss of airway reflexes. Stimulate, bag-valve-mask and suction ready. |

# 1900 Tuesday: Checkpoint 4 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 66, BP 104/60 (MAP 75), RR 16, SpO2 96% room air, temp 97.6, pain 3/10 back and neck catheter site, glucose 54 (at 1905); rhythm: Normal sinus rhythm 66, PR 0.20 s

**Urine for the vitals box:** last hour 20 mL (1500-1900 hourly 15, 20, 20, 20 = 75 mL; plus 500 mL pulled off by dialysis)

## New results (chart values)

| Test | Normal | Tue 1315 | Tue 1700 | Tue 1800 |
|---|---|---|---|---|
| Magnesium | 1.7-2.2 | 10.9 | 7.2 | 4.6 |
| Potassium | 3.5-5.1 | 5.8 | 4.4 | 3.9 |
| Creatinine | 0.7-1.3 | 5.3 | 3.9 | 3.0 |
| Calcium | 8.6-10.2 | 8.3 | 8.0 | 7.7 |
| Ionized calcium | 1.12-1.32 | | | 1.02 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | Dialysis done. Glargine held, but the lispro correction scale every 4 hours is still active: ask to hold it. Low sugar: swallow screen, 15 g carbohydrate by mouth (4 oz apple juice, not orange juice because of potassium), recheck in 15 minutes, repeat, then a snack. Ask for D10W, hourly glucose and a diet order. Vitals and glucose every 15 minutes four times, then hourly four times. |
| **VITALS & ASSESSMENT pair** | Much better: HR 66, BP 104/60, RR 16, SpO2 96% on room air, normal rhythm. But glucose 54, and he is shaky and sweating (awake, oriented, swallowing safely). Pain 3/10 in the back and neck catheter site. Urine 20 mL last hour. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 7.2 (1700) then 4.6 (1800): a big drop, still above normal. Potassium 3.9, creatinine 3.0, calcium 7.7 and ionized calcium 1.02 (both low). ECG 1830: sinus 66, PR 204 ms, QRS 108 ms, QTc 486 (long, from low calcium). Serum glucose was 71 at 1700. |
| **RESULTS & LABS pair: Mg level and strip** | Write 7.2 and 4.6 mg/dL; mark on the strip in the Early band (2.3-5). |
| **TEAM (charge): what changed** | Much better: HR 46 to 66, BP 88/50 to 104/60, RR 9 to 16, off oxygen, magnesium 10.9 to 4.6. New problems after dialysis: glucose 88 to 54, and low calcium. |
| **TEAM (charge): top 3 priorities** | 1) Treat the low glucose safely (swallow screen, 15 g carbohydrate, recheck in 15 minutes) and call the provider. 2) Post-dialysis checks every 15 minutes: BP, HR, rhythm, RR, reflexes, alertness, catheter dressing, tingling. 3) Hand off to the night team: glucose event, calcium and QTc watch, the dialysis catheter, expect magnesium to creep back up, bowel plan. |
| **TEAM (charge): SBAR** | **S:** Mr. Kessler's glucose is 54. He is shaky and sweating, awake and oriented. **B:** 74, kidney disease, dialysis finished at 1800, almost no food for 4 days, long-acting insulin on Monday night. I gave 15 g apple juice after a swallow screen; recheck 68, repeat 15 g, then 84. **A:** Insulin lasts longer with kidney failure, plus dialysis and no food. **R:** Can I have a D10W order, hourly glucose, a plan to hold correction insulin and a diet order? Please review all his insulin. |
| **TEAM (charge): what we will do next** | Treat the glucose by protocol; 15-minute checks; watch calcium symptoms; clear handoff to the night team. |
| **Prediction (no wrong answers)** | Magnesium will probably not stay at 4.6: stored magnesium shifts back into the blood and his kidneys still cannot clear it. What happens: 5.6 at 2100. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** Order D10W with hourly glucose and hold the correction insulin. Give the 15 g by mouth if he passes the swallow screen. Call me if he is under 70 after two rounds.

# 2300 Tuesday night: Checkpoint 5 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 62, BP 112/64 (MAP 80), RR 16, SpO2 96% room air, temp 98.2, pain 3/10 back, glucose 118; rhythm: Sinus rhythm 62, PR 0.22 s

**Urine for the vitals box:** last hour 60 mL (1900-2300 hourly 45, 50, 55, 60 = 210 mL); the guide reads "55 mL/h": accept 45-60

## New results (chart values)

| Test | Normal | Tue 1800 | Tue 2100 |
|---|---|---|---|
| Magnesium | 1.7-2.2 | 4.6 | 5.6 |
| Potassium | 3.5-5.1 | 3.9 | 4.3 |
| Creatinine | 0.7-1.3 | 3.0 | 3.4 |
| BUN | 7-20 | 38 | 40 |
| Calcium | 8.6-10.2 | 7.7 | 7.9 |
| Ionized calcium | 1.12-1.32 | 1.02 | 1.06 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | D10W 40 mL/h running since 1930. The order set has diphenhydramine 25 mg PRN for sleep: do not give it (sedating, can hide slow breathing); ask to stop it and ask for melatonin. Keep gabapentin and oxycodone held; avoid benzodiazepines and zolpidem. Polyethylene glycol and senna due at 2200 are the right kind of bowel plan. No restraints. |
| **VITALS & ASSESSMENT pair** | Newly confused and restless, pulling at lines. Vitals steady: HR 62, BP 112/64, RR 16, SpO2 96% on room air. Glucose 118, so not low. Urine 60 mL last hour. No stool in 4 days. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 5.6 (up from 4.6: rebound), creatinine 3.4 (up), BUN 40, calcium 7.9 and ionized calcium 1.06 (low), potassium 4.3. Possible causes of the confusion: a night in hospital, kidney waste (BUN 40), low calcium, rising magnesium. |
| **RESULTS & LABS pair: Mg level and strip** | Write 5.6 mg/dL; mark on the strip in the Flushing band (5-7). |
| **TEAM (charge): what changed** | Vitals are steady and better than at 1500. New: confusion, restless. Magnesium 4.6 to 5.6, creatinine 3.0 to 3.4. Glucose is back to normal (54 to 118). |
| **TEAM (charge): top 3 priorities** | 1) Safety: protect Ray and his lines (bed alarm, sitter or frequent rounding, secure Foley and neck dressing, quiet room, no restraints). 2) Look for reversible causes and keep NOT giving sedatives; keep RR, reflex and rhythm checks going. 3) Call the night hospitalist (SBAR): stop the sleep-aid order, ask for melatonin, recheck magnesium sooner, bowel plan. |
| **TEAM (charge): SBAR** | **S:** Mr. Kessler is newly confused and restless, pulling at his lines. Vitals are stable and glucose is 118. **B:** Dialysis today for magnesium 10.9. Magnesium 5.6 at 2100 and rising, calcium 7.9. No stool for 4 days. The order set has diphenhydramine, which I am not giving. **A:** I think the confusion comes from the hospital, kidney waste and possibly rebound magnesium. Sedation could hide his breathing and reflexes. **R:** Please stop the diphenhydramine, avoid benzodiazepines, order melatonin, a sitter and delirium precautions, and recheck the magnesium sooner. |
| **TEAM (charge): what we will do next** | Safe, calm room; no sedatives; bowel plan without magnesium; call the provider; call Eleanor. |
| **Prediction (no wrong answers)** | No stool in 4 days and rising magnesium: expect the next level higher and Ray to ask for something for his bowels. What happens: 6.4 at 0100 and "the pink milk". |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** Do not give diphenhydramine; I will stop it and order melatonin and delirium precautions with a sitter. Recheck the magnesium at 0100 (sooner if he gets sleepy), and keep the bowel plan magnesium-free.

# 0300 Wednesday: Checkpoint 6 of 6

**Monitor to set (NOW; give it to the Assessment pair):** HR 56, BP 104/60 (MAP 75), RR 12, SpO2 95% room air, temp 98.0, pain 4/10 abdomen, glucose 126; rhythm: Sinus rhythm 56, PR 0.24 s (long again)

**Urine for the vitals box:** last hour 40 mL (2300-0300 hourly 55, 50, 45, 40 = 190 mL)

## New results (chart values)

| Test | Normal | Tue 1800 | Tue 2100 | Wed 0100 |
|---|---|---|---|---|
| Magnesium | 1.7-2.2 | 4.6 | 5.6 | 6.4 |
| Potassium | 3.5-5.1 | 3.9 | 4.3 | 4.5 |
| Creatinine | 0.7-1.3 | 3.0 | 3.4 | 3.5 |
| Calcium | 8.6-10.2 | 7.7 | 7.9 | 8.1 |
| Ionized calcium | 1.12-1.32 | 1.02 | 1.06 | 1.08 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | D10W stopped at 0130. The bowel protocol (senna, bisacodyl, then milk of magnesia 30 mL if no stool in 72 hours) ends in a magnesium product and disagrees with the no-magnesium order. Do NOT give the pink milk (milk of magnesia); ask to stop the protocol. Safe: polyethylene glycol, senna, bisacodyl suppository once ordered, KUB. No Fleet enema. Calcium only if symptoms or QTc over 500. |
| **VITALS & ASSESSMENT pair** | HR 56, BP 104/60, RR 12, SpO2 95% on room air, reflexes 1+, wakes to voice. PR is longer again (0.24 s). Belly full, pain 4/10 in the abdomen, hypoactive bowel sounds. Urine 40 mL last hour. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 6.4 (0100): the third rise in a row (4.6, 5.6, 6.4), a rebound plus magnesium still in the bowel. Potassium 4.5, creatinine 3.5, calcium 8.1, ionized calcium 1.08. ECG: PR 236 ms (long again). |
| **RESULTS & LABS pair: Mg level and strip** | Write 6.4 mg/dL; mark on the strip in the Flushing band (5-7), close to 7. |
| **TEAM (charge): what changed** | HR 62 to 56, RR 16 to 12, BP 112/64 to 104/60, PR 0.22 to 0.24, magnesium 5.6 to 6.4, new belly pain. Glucose is fine (126). |
| **TEAM (charge): top 3 priorities** | 1) Do not give the milk of magnesia; get the bowel protocol stopped and a magnesium-free bowel plan (bisacodyl, polyethylene glycol, KUB; no Fleet). 2) Watch breathing, reflexes and rhythm every hour until the 0500 level; know the call numbers. 3) Call the provider (SBAR) about the rising magnesium and the abdomen; document the stool. |
| **TEAM (charge): SBAR** | **S:** The 0100 magnesium is 6.4, up from 4.6 after dialysis and 5.6 at 2100. HR 56, BP 104/60, RR 12, reflexes 1+, drowsy. **B:** Dialysis yesterday for magnesium 10.9. No stool in 4 days, belly distended. The bowel protocol ends in milk of magnesia and Ray is asking for it. I have not given it. **A:** Rebound magnesium with magnesium held in the bowel. **R:** Please stop the bowel protocol, order bisacodyl and a KUB, call nephrology about repeat dialysis if magnesium goes over 7, and tell me when to call you again. |
| **TEAM (charge): what we will do next** | No milk; hourly breathing, reflex and rhythm checks; bowel plan; recheck level at 0500; call numbers known. |
| **Prediction (no wrong answers)** | After bisacodyl he should pass stool, and magnesium should start to fall. What happens: large stool at 0345, 5.3 at 0500, alert, reflexes 2+, urine 85 mL/h. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** Do not give the milk of magnesia; I will stop the protocol. Order bisacodyl and a KUB. Recheck magnesium at 0500. Call me and nephrology for RR under 10, HR under 45, SBP under 90, reflexes lost or less alert, or if the magnesium goes over 7.

# 0700 Wednesday: Case closed: final check

**Monitor to set (NOW; give it to the Assessment pair):** HR 68, BP 124/70 (MAP 88), RR 16, SpO2 96% room air, temp 98.0, pain 2/10 back, glucose 112; rhythm: Sinus rhythm 68, PR 0.20 s

**Urine for the vitals box:** last hour 85 mL (0300-0700 hourly 55, 65, 75, 85 = 280 mL)

## New results (chart values)

| Test | Normal | Wed 0100 | Wed 0500 |
|---|---|---|---|
| Magnesium | 1.7-2.2 | 6.4 | 5.3 |
| Potassium | 3.5-5.1 | 4.5 | 4.3 |
| Creatinine | 0.7-1.3 | 3.5 | 3.3 |
| Calcium | 8.6-10.2 | 8.1 | 8.4 |

| Box | Model answer |
|---|---|
| **MEDS & TREATMENTS pair** | Foley can come out; furosemide due at 0800; dialysis catheter stays 24 hours; no more dialysis for now. Keep the no-magnesium rule on the chart and the bowel protocol stopped. Acetaminophen, not ibuprofen, for pain. |
| **VITALS & ASSESSMENT pair** | Alert, reflexes 2+, HR 68, BP 124/70, RR 16, SpO2 96% on room air, pain 2/10 in the back, glucose 112. Urine 85 mL last hour. Large stool at 0345. Weight 90.0 kg. |
| **RESULTS & LABS pair: new results and significance** | Magnesium 5.3 (down from 6.4), creatinine 3.3 (peak 5.3), potassium 4.3, calcium 8.4. ECG 0600: PR 212 ms, QRS 102 ms, sinus 66. |
| **RESULTS & LABS pair: Mg level and strip** | Write 5.3 mg/dL; mark on the strip in the Flushing band (5-7), near the early band. |
| **TEAM (charge): what changed** | Since 0300: alert, reflexes back to 2+, HR 56 to 68, BP 104/60 to 124/70, RR 12 to 16, urine 40 to 85 mL/h, stool passed. Since arrival: magnesium 7.8, peak 10.9, now 5.3. |
| **TEAM (charge): top 3 priorities** | 1) Keep Ray safe from magnesium: no magnesium products at home or in the hospital, read labels, no ibuprofen. 2) Finish the plan: Foley out, furosemide at 0800, watch the dialysis catheter, recheck magnesium and kidney labs. 3) Teach Ray and Eleanor and hand off clearly to the day team. |
| **TEAM (charge): SBAR** | **S:** Mr. Kessler is alert and stable at 0700. Magnesium is down to 5.3 after a rebound to 6.4. **B:** Admitted with magnesium 7.8 from home magnesium products and kidney failure; peak 10.9; dialysis 1500 to 1800; rebound overnight; stool at 0345. **A:** Recovering. Still needs a kidney plan and no magnesium. **R:** Take out the Foley, give furosemide at 0800, keep the catheter 24 hours, recheck labs, and arrange teaching and kidney follow-up. |
| **TEAM (charge): what we will do next** | Foley out; furosemide at 0800; labs; teaching with teach-back; handoff. |
| **What we will teach Ray and Eleanor** | Three plain tips, for example: (1) Do not use milk of magnesia, Maalox-type antacids, magnesium pills or Epsom salts; look for the word magnesium on the label. (2) For constipation, heartburn, cramps or pain, call the kidney doctor or pharmacist first (ask about MiraLAX or senna, and acetaminophen instead of ibuprofen). (3) Bring every bottle to every visit and say "Ray has kidney disease"; call before giving his blood pressure or water pill when he is vomiting. |
| **The story of the stay** | 7.8 on arrival, 8.7 at 0900, peak 10.9 at 1315, 7.2 and 4.6 with dialysis, then back up to 5.6, 6.4 and finally 5.3. |
| **Charge nurse notes** | Look for: each pair was asked, the charge nurse decided, and a time check. Credit any honest notes. |

**Instructor as the provider (suggested reply):** Good work. Take out the Foley, furosemide at 0800, keep the dialysis catheter another 24 hours, recheck labs this morning and plan teaching and kidney follow-up.

## Scoring guide (20 points, three levels)

Give ONE group score from the shared packet (group score sheet, next page) plus an individual check for each of the 7 students. Be generous: complete and thoughtful earns full credit. Individual reflections and peer checks feed two of the rows.

| Category | Full credit | Good | Getting there |
|---|---|---|---|
| **Teamwork and charge-nurse rotation (5)** | 5. Every section has a charge nurse named, everyone was charge nurse once, and each pair reported its job at every checkpoint. | 3 to 4. The rotation was followed most of the time. Most teammates took part. | 1 to 2. The rotation was skipped, or one person did most of the work. |
| **Clinical thinking at each checkpoint (6)** | 6. Every checkpoint has vitals, meds, results and changes filled in, with real numbers from the chart and a reason. | 4 to 5. Most checkpoints are filled in with numbers. One or two are thin. | 1 to 3. Many boxes are blank or have one-word answers. |
| **Safety, priorities and SBAR (5)** | 5. At every checkpoint, 3 priorities in order, a complete SBAR (all four parts) and a next step that keeps Ray safe. | 3 to 4. Most checkpoints have priorities and an SBAR. A part is sometimes missing. | 1 to 2. Priorities or SBAR are missing at many checkpoints. |
| **Completeness and reflection (4)** | 4. All boxes tried, the magnesium trend strip is marked, and each person handed in the reflection page. | 2 to 3. Most boxes tried. The strip or a reflection is missing. | 1. Many blanks. Little reflection. |

### How the points add up

| Category | Where the points come from |
|---|---|
| Teamwork and charge-nurse rotation (5) | Charge nurse named for all 7 sections = 2. Everyone had a turn as charge = 1. Whole team worked at the checkpoints and handed in the peer check = 2. |
| Clinical thinking at each checkpoint (6) | 1 point for each of the 6 checkpoints (0700, 1100, 1500, 1900, 2300, 0300). Half a point is fine for a mostly complete page. |
| Safety, priorities and SBAR (5) | Priorities and SBAR complete at 5 or 6 checkpoints = 5; at 4 = 4; at 3 = 3; at 2 = 2; at 1 = 1. |
| Completeness and reflection (4) | Boxes tried = 2. Trend strip marked = 1. Individual reflections handed in = 1. |

### Reflection and peer check

- Reflection (3 boxes): full credit for any honest answer with a reason. A blank or one-word box is a partial credit. Use it for the "Individual reflections handed in" point.
- Peer check: use it to spot a team where one person did most of the work. Read the comments privately; do not share names.
- If the team did not rotate the charge nurse, cap Teamwork at 3 and ask why in the debrief.

# GROUP SCORE SHEET

Team name: _____ Date: _____ (one group score, then 7 individual checks)

## Checkpoint check (tick what you see)

| Page | Charge nurse named | Pair boxes done with numbers | 3 priorities | SBAR complete | Mg on strip | Prediction |
|---|---|---|---|---|---|---|
| 0700 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 1100 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 1500 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 1900 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2300 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 0300 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Final 0700 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

## Rubric

| Category | Full | Good | Getting there | Points | Notes |
|---|---|---|---|---|---|
| Teamwork and charge-nurse rotation (5) | ☐ 5 | ☐ 3-4 | ☐ 1-2 | / 5 | |
| Clinical thinking at each checkpoint (6) | ☐ 6 | ☐ 4-5 | ☐ 1-3 | / 6 | |
| Safety, priorities and SBAR (5) | ☐ 5 | ☐ 3-4 | ☐ 1-2 | / 5 | |
| Completeness and reflection (4) | ☐ 4 | ☐ 2-3 | ☐ 1 | / 4 | |
| TOTAL | | | | / 20 | |

## Individual check

| Student (letter, pair) | Was charge nurse (1) | Reflection (2) | Peer check (1) | Individual score /4 |
|---|---|---|---|---|
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |
| | ☐ | ☐ | ☐ | |

**Comments for the team:**

| |
|---|
| |
| |

## Debrief questions (about 10 minutes; pick 8 to 10)

| # | Topic | Question |
|---|---|---|
| 1 | Charge nurse | What did the charge nurse do that helped the team most? What was hard about coordinating? |
| 2 | Charge nurse | How did the team decide who looked at what? Did that change from one checkpoint to the next? |
| 3 | Trend | Look at the Mg strip. Tell the story of Ray's stay in one minute. When was the most dangerous point and how did the team know? |
| 4 | Cause | Where did the magnesium come from, and why could Ray not get rid of it? Which medicines hurt his kidneys? |
| 5 | Safety | At 1100 Ray was on oxygen with a normal-looking SpO2. Why can a good SpO2 be misleading when magnesium is high? |
| 6 | Safety | Which orders could have hurt Ray (bowel protocol, sleep order, correction insulin, the milk of magnesia request)? How did the team catch them? |
| 7 | Dialysis | Why did magnesium rise again after dialysis? What other problems can appear after dialysis? |
| 8 | SBAR | Which SBAR call was the clearest? What made it clear? What would you add to a weaker one? |
| 9 | Prediction | Which prediction came closest? What clue would help you predict the next change earlier? |
| 10 | Teaching | What would you tell Ray and Eleanor so this does not happen again? |

### Ideas for good answers

- Charge nurse: said who looks at what out loud; watched the clock; made sure the Mg was marked; kept the SBAR short.
- Trend: 7.8, 8.7, 9.5, 10.9 at the peak, dialysis to 4.6, then 5.6, 6.4 and 5.3. Rebound happens because stored magnesium shifts back and the kidneys cannot clear it.
- Oxygen: SpO2 can look fine while breathing is slow and shallow; watch RR, EtCO2 and reflexes.
- Orders that could hurt: the bowel protocol ends in milk of magnesia; diphenhydramine for sleep; correction insulin after dialysis; "pink milk".
- Teaching: read labels for magnesium; call before laxatives or antacids; no ibuprofen; bring all bottles; sick-day rule for the water pill and blood pressure pill.